Optimisation landscape

Problem /Method	Algorithm Family	Software package[1] Opensource Commercial	Optimization reach Optimization-only Decision-integrated	Characteristics
1. MILP - Single/multi objectives - **Linear objective** functions - (Conflicting) objectives - Mixed-integer variables - **Feasibility** constraints	**Exact/ linear programming (LP)** : Simplex and/or Branch-and-Bound (-Cut). MOO via different methods, e.g.: WAM performance aggregation + (optional) Pareto-weights; ε-constraint ; distance-based formulation (ideal-point).	Opensource: ORTools, HiGHS, SCIP, CBC, Choco-solver Commercial: Gurobi, CPLEX, COPT, FICO Xpress	**MOO:** multi-objective optimisation for performance. Requires a-posteriori MCDA for evaluation; no direct best-fit	Highly scalable and fast. Strictly linear structure ⇒ global optimality guaranteed. Often not suitable for complex real-world engineering and scheduling problems. Mainly suitable for teaching and introductory use.
2. MINLP (MIP) - Single/multi objectives - **Non-linear objective** functions - (Conflicting) objectives - Mixed-integer variables - **Feasibility** constraints	**Quasi-exact/ constraint programming (CP)**: decomposition-based ; specific hybrid or transformed MILPs.	Opensource: ORTools, SCIP Commercial: BARON, AntiGone, Gurobi*, CPLEX*	**MOO:** multi-objective optimisation for performance. Requires a-posteriori MCDA for evaluation; no direct best-fit	Rigid; problem-class specific; requires exact math formulation. Guarantees objective optimality, but not decision-best-fit.
	Evolutionary / Pareto heuristics: Genetic Algorithms: e.g., NSGA-II, MOEA/D, SPEA, BRKGA.	Opensource: Pymoo, DEAP	**MOO:** multi-objective Pareto optimality. Requires a-posteriori MCDA for evaluation; no direct best-fit	Black-box; Pareto methods cover only part of real-world problems. No guarantee of decision-best-fit (dominance in objective space only).
3. IMAP - Single/multi performance - **(Non)-linear** system **performance** functions - **(Non)-monotonic** multi-actor **preference** functions - (Conflicting) preferences - All type of variables - **Feasibility & acceptability** constraints	**Intergenerational / Preference-based:** Population based metaheuristic: IMAP-IGS (instantiations: IGS-GA I, IGS-GA II IGS-BRKGA, IGS-SI).	Opensource: PREFERENDUS	Decision-integrated: **MODO:** multi-objective design/decision optimisation, Directly solves for 'best-fit' design-decision point; pure support for multi-criteria design & decision-making (MCDM)	Design/decision-valid: integrates performance and preferences. Preferences explicitly modeled from performance and/or attributes. Yields a best-fit solution in aggregated weighted-preference design-decision space. Flexible for highly constrained, multi-actor real-world problems (incl. preference-based constraint handling). Also able to solve (non)linear SODOs.

[1] Among the most relevant
* Limited (mainly quadratic)